The regular n-gon of point vortices is stable at its bifurcation point: Montaldi–Tokieda's Conjecture 13 for every $n \geq 4$

Gordon Speagle
Independent Researcher, Boise, ID, USA
gspeagle@gmail.com

Draft, 23 October 2026

Abstract

Montaldi and Tokieda put point vortices on the sphere, the plane and the hyperbolic plane into one family of geometries and showed that a regular ring of $n \geq 4$ identical vortices loses stability at an explicit value b_n of the curvature parameter. They conjectured that the ring is still stable *at* b_n (their Conjecture 13). We prove this for every $n \geq 4$. At b_n the reduced Hamiltonian has a degenerate critical point whose Hessian is positive semidefinite with a one- or two-dimensional kernel, so stability is decided by a quartic coefficient c_4. We compute c_4 on the momentum level set, in closed form for all n:

$$c_4 = \frac{(n-2)\,(m^4 + 40m^2 + 48m + 16)}{192\,n(n-1)} \quad (n \text{ even},\ m = n-2),$$
$$c_4 = \frac{(n-1)(n-3)(n^2 - 9n + 38)}{128\,n} \quad (n \text{ odd}),$$

both positive. A standard energy–momentum argument for degenerate minima then gives Lyapunov stability modulo rotations. Working on the level set matters: the level set bends at second order into a direction where the Hessian is not zero, and this bending contributes at fourth order. For even n, Montaldi and Tokieda expand along a straight line that leaves the level set. Their quartic has the right sign but not the right value: at $n = 4$ the t^4 coefficient along their line is $6 - 4\sqrt{3}/3 \approx 3.69$, while the level set gives 4. The algebra and the series coefficients (the closed forms, positivity, the Taylor coefficients derived from the Hamiltonian, the root-of-unity sums, the Fourier blocks of the Hessian, the comparison with the straight line), and the abstract Lyapunov lemmas, are checked in Lean 4. Reducing the problem to these statements, and instantiating the abstract argument for the vortex flow, are done by hand. No human specialist has refereed the proof.

Contents

1 Introduction **2**

2 Setting, threshold and spectrum **3**
2.1 The family 3
2.2 The ring and its threshold 4
2.3 Fourier blocks 4
2.4 The reduced space and the kernel 5

3 The quartic coefficient on the momentum level set **6**

4 Even n **7**

5 Odd n **8**

6 Comparison with Montaldi–Tokieda's even-n computation **10**

7 Nonlinear stability at a degenerate minimum **11**

8 A direct integration **14**

9 Relation to prior work **15**

10 Status and limitations **15**

A Coefficients for odd n **17**

B Expansions behind h_4 and the E_{ij} **17**

1 Introduction

The question. A ring of n identical point vortices placed at the vertices of a regular polygon rotates rigidly. Whether this rotating ring is stable is one of the oldest questions in vortex dynamics: Thomson [14] found that in the plane the ring is linearly stable for $n \leq 6$, unstable for $n \geq 8$, and degenerate for $n = 7$. The heptagon was shown to be nonlinearly stable much later [6, 13]. On a sphere the answer depends on latitude, and on the hyperbolic plane on the radius.

Montaldi and Tokieda [8] ("MT") put the sphere, the plane and the hyperbolic plane into a single one-parameter family of geometries. In it the ring's stability is governed by one number, $b = \lambda r_0^2$ (curvature times squared radius, in their normalisation). They prove that for $n \geq 4$ the ring is stable for $b < b_n$ and unstable for $b > b_n$, where b_n is explicit (equation (2.2) below), and that a pitchfork bifurcation occurs at b_n. The threshold b_n lies on the sphere for $n = 4, 5, 6$, is the plane ($b_7 = 0$, Thomson's heptagon) for $n = 7$, and lies on the hyperbolic plane for $n \geq 8$. At the bifurcation point itself the linearisation is degenerate, so stability is a nonlinear question. MT state ([8, p. 23], arXiv:1210.5662v1):

> **Conjecture 13.** *For all values of n, the n-gon is stable at the bifurcation point.* Above, we have proved this for all even n and for $n = 3, 5, 7, 9, 11$.

Here "stable" means Lyapunov stability modulo the rotations about the ring's centre [8, pp. 10, 15]. MT treat odd $n = 5, 9, 11$ by floating-point evaluation of three coefficients, $n = 7$ by computer algebra, and even n by a closed-form computation of a quartic coefficient. The odd cases $n \geq 13$, which all lie on the hyperbolic plane, are open in [8].

Result.

Theorem 1.1 (Main theorem)**.** *Let $n \geq 4$ and consider n identical point vortices in MT's family at the bifurcation point $\lambda r_0^2 = b_n$ (sphere for $n = 4, 5, 6$, plane for $n = 7$, hyperbolic plane for $n \geq 8$). Let $K \cong SO(2)$ be the rotations about the centre of the regular ring x_0. Then $K \cdot x_0$ is Lyapunov stable: for every $\varepsilon > 0$ there is $\delta > 0$ such that every solution with $\operatorname{dist}(x(0), K \cdot x_0) < \delta$ exists for all $t \in \mathbb{R}$ and satisfies $\operatorname{dist}(x(t), K \cdot x_0) < \varepsilon$. No restriction is placed on the momentum of $x(0)$.*

Here dist *is the Euclidean distance of labelled configurations* $(z_1, \dots, z_n) \in \mathbb{C}^n$ *in MT's chart; since* $K \cdot x_0$ *is compact, any metric inducing the same topology gives an equivalent statement.*

So Conjecture 13 holds for every $n \geq 4$. (The case $n = 3$ is different; see Remark 5.2.)

The idea. By the energy–momentum method, it is enough to show that the reduced Hamiltonian h_μ on the reduced space $J^{-1}(\mu)/K$ has a strict local minimum at the ring. At b_n the Hessian of h_μ is positive semidefinite with kernel the critical Fourier mode (one-dimensional for even n, two-dimensional for odd n). Symmetry excludes a cubic term. So the decisive quantity is the quartic coefficient c_4 of h_μ along the kernel, after the remaining directions are minimised out. We compute c_4 exactly for every n (Theorems 4.1 and 5.1) and find it positive. A degenerate version of the energy–momentum theorem (Section 7) turns this into Theorem 1.1.

Why the level set matters. At a nondegenerate minimum, one may compute on the linear slice $x_0 + \ker dJ(x_0)$ instead of on the curved level set $J^{-1}(\mu)$; the two differ at second order in the displacement, which is invisible next to the quadratic term. At a degenerate minimum this is no longer true. A curve in $J^{-1}(\mu)$ tangent to the kernel has a second-order component along the radial mode V_0, which is not in $\ker dJ$ and on which the Hessian is nonzero. This component contributes at fourth order, which is the leading order. MT's even-n computation [8, §3.5.1] expands along the straight line $z_j = (r_0 + (-1)^j t)e^{2\pi i j/n}$, which lies in the linear slice but leaves the level set. Section 6 computes both quartics exactly. They are both positive, but they differ for every n we tested; at $n = 4$ the line gives $6 - 4\sqrt{3}/3 \approx 3.69$ and the level set gives 4 (Figure 1). So MT's even-n argument does not by itself establish the even case, although the even case is true.

What is checked how. Each claim below carries a status label: [Proved] (hand proof here), [Lean] (checked in Lean 4 with Mathlib; we name the module, then the theorems; full names are in the README), [Certified] (exact computer algebra; script named), [Numerical (exact modular)] (exact arithmetic modulo large primes: strong evidence, not a proof over $\mathbb{Q}$), [Numerical] (floating point). The paper's directory contains the scripts, their outputs, and a table mapping every claim to its check. In short: Lean checks the arithmetic of the threshold and of the spectral computation, the Taylor coefficients of the Hamiltonian along the curves used (derived from the Hamiltonian itself), the root-of-unity sums, the closed forms and positivity of c_4 for every n, the comparison with MT's straight line, the Fourier blocks of the Hessian at the ring, the bracket $\{J_1, J_2\}$, and the abstract Lyapunov lemmas. The structural reductions (the reduced space, symmetric criticality, completeness of the odd ansatz, and the instantiation of the abstract stability argument for the vortex flow) are by hand. Section 10 lists this precisely.

Plan. Section 2 sets up the family, the threshold and the spectral picture. Section 3 defines c_4 on the level set. Sections 4 and 5 compute it for even and odd n. Section 6 compares with MT's straight line. Section 7 gives the nonlinear argument and proves Theorem 1.1. Section 8 shows a direct integration, Section 9 discusses prior work, and Section 10 states the status and limitations.

2 Setting, threshold and spectrum

2.1 The family

Following [8], in a chart $z \in \mathbb{C}$ put $\sigma(z) = 1 + \lambda|z|^2$. The area form $\Omega_\lambda = 2\sigma^{-2}\, dx \wedge dy$ [8, (1.8)] is that of a surface of constant curvature 4λ: the round sphere in stereographic coordinates for $\lambda > 0$,

the plane for $\lambda = 0$, and the hyperbolic plane for $\lambda < 0$ (the disk $|z|^2 < -1/\lambda$). For n identical unit vortices at distinct points $z_1, \dots, z_n$ we use

$$H = -\sum_{i<j} \log \frac{|z_i - z_j|^2}{\sigma_i \sigma_j}, \qquad \Omega = \sum_k \frac{2}{\sigma_k^2}\, dx_k \wedge dy_k, \qquad J = \sum_k \Big(\frac{x_k}{\sigma_k}, \frac{y_k}{\sigma_k}, \frac{|z_k|^2}{\sigma_k} \Big), \tag{2.1}$$

with $\sigma_k = \sigma(z_k)$. This is MT's Hamiltonian [8, (1.12)] without the factor $\kappa^2/4\pi$, which only rescales time, and J is their momentum map [8, (1.9)]. For $\lambda < 0$ the pair interaction is $-\log \sinh^2(d/2)$ up to constants (d the hyperbolic distance), the "sinh" model. The symmetry group is $SO(3)$, $SE(2)$ or $PSL(2,\mathbb{R})$; J_3 generates the rotations K about $z = 0$.

Remark 2.1 (Normalisation). All quartic coefficients in this paper refer to (2.1). With the common alternative $H' = -\sum_{i<j} \log d_{ij} = \frac{1}{2} H + \text{const}$ (d_{ij} the chordal or sinh distance), every quartic coefficient is halved: $c_4' = \frac{1}{2} c_4$, so the closed forms have denominators $384\, n(n-1)$ and $256\, n$. Some of the copied scripts print c_4'; the paper's README says which.

2.2 The ring and its threshold

The regular ring $z_k = r_0 e^{2\pi i k/n}$ is a relative equilibrium for every r_0 and λ [8, (2.1)]. By scaling we take $r_0 = 1$, so the only parameter is $b = \lambda \in (-1, 1]$ (the constraint $b > -1$ keeps the ring inside the hyperbolic disk). MT [8, Thm. 6] prove that for $n \geq 4$ the ring is stable for $b < b_n$ and spectrally unstable for $b > b_n$, where b_n solves

$$\frac{1 + b_n^2}{(1 + b_n)^2} = \frac{1}{2(n-1)} \Big\lfloor \frac{n^2}{4} \Big\rfloor \qquad \text{[8, (3.6)]}. \tag{2.2}$$

Proposition 2.2 (Threshold). *For every $n \geq 4$, (2.2) has exactly one root b_n in $(-1, 1]$, and $-1 < b_n < 1$. Explicitly $b_n = (4(n-2)\sqrt{n-1} - n^2)/(n^2 - 8n + 8)$ for even n, $b_n = (4\sqrt{n-3} - (n+1))/(n-7)$ for odd $n \neq 7$, and $b_7 = 0$. Moreover $b_n > 0$ for $n \leq 6$ and $b_n < 0$ for $n \geq 8$; for example $b_4 = 2 - \sqrt{3}$, $b_6 = 9 - 4\sqrt{5}$, $b_8 = 3\sqrt{7} - 8$, $b_{10} = -1/7$.*

[Lean] `Conj13.Algebra`: `bn_spec`, `bn_unique`, `bn_even_paper`, `bn_odd_paper`, `bn_seven`, `bn_sign`, `bn_four` ... `bn_ten`. [Certified] `code/threshold.py`.

2.3 Fourier blocks

Write $z_k = e^{s_k + i(\phi_k + \theta_k)}$ with $\phi_k = 2\pi k/n$, so the ring is $s = \theta = 0$. Let $F = H - \omega_0 J_3$, with ω_0 chosen so that $dF(x_0) = 0$ (the ring rotates with angular velocity proportional to ω_0). In these coordinates

$$\log |z_j - z_k|^2 = s_j + s_k + \log \big(2\cosh(s_j - s_k) - 2\cos(\phi_j - \phi_k + \theta_j - \theta_k) \big)$$

does not involve λ; all dependence on λ is in the one-body terms $(n-1)\sum_k \log(1 + b e^{2s_k})$ in H and $J_3 = \sum_k \mathcal{G}(e^{2s_k})$, $\mathcal{G}(u) = u/(1+bu)$. Let V_m ($0 \leq m \leq \lfloor n/2 \rfloor$) be the real Fourier spaces spanned by $(\cos m\phi_k)_k$ and $(\sin m\phi_k)_k$ in s and in θ (dimension 4 for $0 < m < n/2$, 2 for $m = 0$ and $m = n/2$).

Proposition 2.3 (Fourier blocks, [8, (2.6)]). *For $b \in (-1, 1)$, $\omega_0 = -(n-1)(1-b^2)/2$, and $d^2F(x_0)$ is block-diagonal over the V_m, with no radial–tangential coupling. With respect to $\sum_k (ds_k^2 + d\theta_k^2)$ its eigenvalues on V_m are*

$$\alpha_m = C_1 - m(n-m) \quad \textit{(radial)}, \qquad \beta_m = m(n-m) \quad \textit{(tangential)}, \qquad C_1 = \frac{2(n-1)(1+b^2)}{(1+b)^2}.$$

At $b = b_n$ one has $C_1 = \lfloor n^2/4 \rfloor = \max_m m(n-m)$; this is (2.2).

Proof. The pair logarithm has no linear term in $s_j - s_k$ and no mixed $s\theta$ term at second order; its quadratic part in the differences is $(\Delta s^2 - \Delta\theta^2)/(2 - 2\cos\Delta\phi)$. On a Fourier mode the pair sum is diagonalised by the identity

$$\sum_{l=1}^{n-1} \frac{1-\cos(2\pi ml/n)}{1-\cos(2\pi l/n)} = m(n-m) \qquad (0 \le m \le n), \tag{2.3}$$

(with $w = e^{2\pi il/n}$ the summand is $|1 + w + \cdots + w^{m-1}|^2$). The one-body part is $\sum_k \varphi(s_k)$ with $\varphi(s) = -(n-1)s + (n-1)\log(1+be^{2s}) - \omega_0 e^{2s}/(1+be^{2s})$; then $\varphi'(0) = 0$ fixes ω_0 and $\varphi''(0) = C_1$. □

[Proved]. [Lean], from the Hamiltonian (2.1) as a function of the $2n$ coordinates (s, θ), for every $b > -1$ (sphere, plane and hyperbolic plane at once): `Conj13.MTBlocks`: `mt_grad_formula` and `mt_ring_critical_iff` (the ring is critical exactly for $\omega = \omega_0$), `mtD_omega` (one-body diagonal $= C_1$), `mtHessQ_eq` (the Hessian form as explicit circulant matrices), `mt_blocks_eigen` and `mt_blocks_hermitian` (on each Fourier vector $(w^{mc})_c$ the blocks are α_m, β_m and 0, and different modes do not couple). Here the Hessian is $t \mapsto d^2F(x_0 + t\,\cdot)/dt^2$ at 0 and its entries are obtained by polarisation; the identification with the matrix of second partial derivatives is the standard fact for C^2 functions and is not restated in Lean. The pair-term calculus and the Fourier sums are reused from a companion Lean development (`Conj13.Ring`); `Conj13.Algebra`: `fejer`, `phi_crit_and_C1`, `C1_eq_iff` ($C_1 = \lfloor n^2/4 \rfloor \Leftrightarrow$ (2.2)). [Numerical] `code/spectral_check.py` compares the full $2n \times 2n$ Hessian, compressed to $\ker dJ \cap t^\perp$, with these predictions for $n = 4, \ldots, 15$ at 40 digits.

2.4 The reduced space and the kernel

Let $\mu = J(x_0) = (0, 0, \mu_3)$. Because $|b_n| < 1$, the ring is not on the equator of the sphere, so the stabiliser G_μ of μ is K. Let $t = (0, \ldots, 0; 1, \ldots, 1)$ be the rotation direction ($\theta_k = 1$).

- dJ_3 is nonzero exactly on the radial part of V_0.
- $J_+ = J_1 + iJ_2$ satisfies $J_+ \circ R = e^{2\pi i/n} J_+$ for the symmetry R (rotation by $2\pi/n$ with relabelling), so $dJ_+(x_0)$ vanishes off V_1. It has real rank 2 on V_1; put $V_1' = V_1 \cap \ker dJ_+$.
- $\{J_1, J_2\}(x_0) = n(1-b)/(2(1+b)) \neq 0$. (For $\lambda = 0$, $\{J_1, J_2\} = n/2$ is the central term of the plane's non-equivariant momentum map; we never use equivariance.)

Hence $\ker dJ(x_0) = \mathbb{R}t \oplus V_1' \oplus V_2 \oplus \cdots \oplus V_{\lfloor n/2 \rfloor}$, the reduced space $M_\mu = J^{-1}(\mu)/K$ is a manifold of dimension $2n - 4$ near $[x_0]$ (K acts freely on labelled configurations), and on $J^{-1}(\mu)$ we have $H = F + \omega_0\mu_3$.

Proposition 2.4 (Spectral picture at the threshold)**.** *Let $n \ge 4$ and $b = b_n$. The Hessian of the reduced Hamiltonian h_μ at $[x_0]$ is $d^2F(x_0)$ restricted to $\ker dJ(x_0)$ modulo $\mathbb{R}t$. It is positive semidefinite with kernel Z_0 equal to the radial part of $V_{\lfloor n/2 \rfloor}$ (dimension 1 for even n, 2 for odd n), and positive definite on the complement*

$$E = V_1' \oplus \bigoplus_{2 \le m \le \lfloor n/2 \rfloor} (\textit{tangential part of } V_m) \oplus \bigoplus_{2 \le m < \lfloor n/2 \rfloor} (\textit{radial part of } V_m).$$

Proof. For a curve c in $J^{-1}(\mu)$ through x_0, $(F \circ c)''(0) = d^2F(c', c')$ since $dF(x_0) = 0$. By Proposition 2.3 the restriction is block-diagonal. On V_1, $d^2F = \mathrm{diag}(\alpha_1, \beta_1)$ with $\beta_1 = n - 1 > 0$ and $\alpha_1 = \lfloor n^2/4 \rfloor - (n-1) > 0$ for $n \ge 4$, so it is positive on V_1'. For $m \ge 2$, $\beta_m > 0$, and $\alpha_m = (\lfloor n/2 \rfloor - m)(\lfloor n/2 \rfloor - m + (n \bmod 2)) \ge 0$ with equality iff $m = \lfloor n/2 \rfloor$. The radial part of V_0 is removed by $J_3 = \mu_3$ and its tangential part is $\mathbb{R}t$. □

[Proved]; the integer arithmetic is [Lean] `Conj13.Algebra`: `am_factor`, `am_nonneg_and_kernel`, `am_next` (the next eigenvalue is 1 for even and 2 for odd n), `V1'_form`, `V1'_n4`, `V1'_n5` (the V_1' eigenvalue is $3/2$ at $n = 4$ and $8/3$ at $n = 5$), `bracket_value` (the value at the ring); `Conj13.Bracket`: `hasDerivAt_J1_x`, ..., `bracket_one`, `bracket_sum` (the partial derivatives of $J_1 = x/\sigma$, $J_2 = y/\sigma$ and $\{J_1, J_2\} = \sum_k (\frac{1}{2} - \lambda|z_k|^2/\sigma_k) = \frac{n}{2} - \lambda J_3$, taking as definition the one-vortex Poisson bracket $\{f, g\} = \frac{\sigma^2}{2}(f_x g_y - f_y g_x)$ of $\Omega = 2\sigma^{-2} dx \wedge dy$).

3 The quartic coefficient on the momentum level set

Let $e \in \mathbb{R}^{2n} \cong \mathbb{C}^n$ be the Euclidean-unit radial kernel vector in the chart: $e_k = n^{-1/2}(-1)^k \hat{z}_k$ for even n, and $e_k = (2/n)^{1/2} \cos(l\phi_k)\hat{z}_k$, $l = (n-1)/2$, for odd n, where $\hat{z}_k = e^{i\phi_k}$. For odd n let e' be the same with sin in place of cos. Let t_x be the rotation tangent at x_0 (in the chart, $(t_x)_k = i\hat{z}_k$).

Definition 3.1 (Level-set quartic)**.** For small ε let

$$f(\varepsilon) = \min\big\{H(q) - H(x_0) :\ J(q) = \mu,\ \langle q - x_0, e\rangle = \varepsilon,\ \langle q - x_0, e'\rangle = 0\ (n \text{ odd}),\ \langle q - x_0, t_x\rangle = 0,\ q \text{ near } x_0\big\},$$

and $c_4 = \lim_{\varepsilon \to 0} f(\varepsilon)/\varepsilon^4$.

The constraint $\langle q - x_0, t_x\rangle = 0$ is a local section of the K-action on $J^{-1}(\mu)$, and $q \mapsto (\langle q - x_0, e\rangle, \langle q - x_0, e'\rangle, (\text{projection onto } E))$ restricted to that section is a chart of M_μ near $[x_0]$ whose linear part is the splitting $Z_0 \oplus E$. In this chart write $h_\mu = h_\mu(a, y)$ with $a \in Z_0$, $y \in E$. By Proposition 2.4 and the implicit function theorem there is a unique smooth $y^*(a) = O(|a|^2)$ with $\partial_y h_\mu(a, y^*(a)) = 0$, and $y \mapsto h_\mu(a, y)$ is strictly convex near $y^*(a)$. So

$$g(a) := h_\mu(a, y^*(a)) - h_\mu(0) = \min_y h_\mu(a, y) - h_\mu(0),$$

and $f(\varepsilon) = g(\varepsilon, 0)$. The function g is the reduced Hamiltonian with the stable directions minimised out; its Taylor series starts at order 4 (below).

Lemma 3.2 (Chart independence)**.** *Let g_1, g_2 be defined as above in two charts of M_μ whose linear parts induce the same projection onto Z_0 along E, up to a linear isomorphism L of Z_0. If $g_1(a) = q_1(a) + O(|a|^5)$ with q_1 a quartic form, then $g_2(a) = q_1(L^{-1}a) + O(|a|^5)$. In particular the sign of the quartic term does not depend on the chart.*

Proof. Let $x_1(a)$ be the minimiser on the chart-1 fibre over a. It lies on the chart-2 fibre over $a' = La + O(|a|^2)$, because $x_1(a) = (a, O(|a|^2))$ in chart 1 and the transition map is linear plus quadratic. So $g_2(La + O(|a|^2)) \le g_1(a)$, and symmetrically. Since both functions are $O(|a|^4)$ with C^5 remainders, a shift of the argument by $O(|a|^2)$ changes them by $O(|a|^5)$, and the two inequalities give the claim. □

[Proved]. Lemma 3.2 lets us compute in convenient coordinates: the even-n computation uses radii of two polygons, the odd-n computation uses canonical coordinates, and both are normalised to the unit vector e.

Lemma 3.3 (Symmetry)**.** *g has no quadratic and no cubic term, and its quartic term is $c_4|a|^4$ (for odd n, with $|a|$ the Euclidean norm on $Z_0 \cong \mathbb{R}^2$ given by e, e').*

Proof. The quadratic term vanishes by Proposition 2.4. The function g is invariant under the $\mathbb{Z}_n$ generated by R, which preserves h_μ, J, the splitting and the section. For even n, R acts on $Z_0 = \mathbb{R}e$ by -1, so g is even. For odd n, R acts on $Z_0 \cong \mathbb{C}$ by multiplication by $\zeta = e^{2\pi i l/n}$, a primitive n-th root of unity since $\gcd(l, n) = 1$. A monomial $a^p \bar{a}^q$ is invariant iff $n \mid p - q$; for $p + q \le 4 < n$ this forces $p = q$. So there is no cubic term and the only quartic is $|a|^4$, as in [8, Prop. 12]. The reflection fixing vortex 0 maps $e \mapsto e$ and $e' \mapsto -e'$ and is consistent with this. □

[Proved]; the arithmetic is [Lean] `Conj13.Symmetry`: `invariant_monomial` ($n \ge 5$, $p + q \le 4$, $n \mid p - q \Rightarrow p = q$), `crit_mode_coprime` ($\gcd(M, 2M + 1) = 1$).

Remark 3.4 (The quartic sees the curvature of the level set). The quartic of g is built from three pieces: the fourth derivative of F along the kernel, the cubic coupling of a^2 to the slaved $O(|a|^2)$ displacement, and the quadratic energy of that displacement. On the level set, the displacement includes the $O(|a|^2)$ bending of $J^{-1}(\mu)$ out of the linear slice, chiefly along radial V_0, where d^2F is not zero ($\alpha_0 = C_1$). Its contribution is of order $|a|^4$: the same order as the quantity being computed. So c_4 must be computed on $J^{-1}(\mu)$, not on $x_0 + \ker dJ(x_0)$. Section 6 makes this explicit.

4 Even n

Theorem 4.1. *For even $n = 2M \ge 4$ at $b = b_n$,*

$$c_4 = \frac{(n-2)\,(m^4 + 40m^2 + 48m + 16)}{192\,n(n-1)}, \qquad m = n - 2.$$

In particular $c_4 > 0$, and $c_4 = \frac{1}{4}, \frac{23}{30}, \frac{95}{56}, \frac{49}{15}, \frac{755}{132}$ for $n = 4, 6, 8, 10, 12$.

Proof. Symmetric criticality. Let $D_M \subset D_n$ be generated by rotation through $2\pi/M$ (two steps) and the reflection $z \mapsto \bar{z}$, $k \mapsto -k$. Both fix e (they preserve the parity of k) and fix x_0; the rotation fixes t_x and the reflection maps it to $-t_x$, so all constraints in Definition 3.1 are D_M-invariant, as is H. For small ε the minimisation defining $f(\varepsilon)$ is strictly convex in a fixed neighbourhood (Proposition 2.4), so its minimiser is unique and therefore D_M-invariant. Near the ring, the D_M-fixed labelled configurations are exactly two staggered regular M-gons,

$$\{\rho_+ \eta^j\}_{j<M} \cup \{\rho_- \zeta \eta^j\}_{j<M}, \qquad \eta = e^{2\pi i/M},\ \zeta = e^{i\pi/M},$$

with $\rho_\pm$ real and near 1. On them $J_1 = J_2 = 0$ and $\langle q - x_0, t_x \rangle = 0$ automatically, and $\langle q - x_0, e \rangle = \frac{\sqrt{n}}{2}(\rho_+ - \rho_-)$ exactly. The remaining constraint $J_3 = \mu_3$ cuts out a curve. So $f(\varepsilon)$ is the value of $H - H(x_0)$ on this curve.

Product formula. With $u_\pm = \rho_\pm^2$ and $\prod_{\omega^M=-1}(x - y\omega) = x^M + y^M$,

$$H = -\tbinom{M}{2}(\log u_+ + \log u_-) - 2M \log\left(u_+^{M/2} + u_-^{M/2}\right) + (2M-1)M\left(\log(1+bu_+) + \log(1+bu_-)\right) + \text{const.}$$

Expansion on the constraint. Parametrise $J_3 = \mu_3$ by $\mathcal{G}(u_\pm) = \mathcal{G}(1) \pm \tau$. The swap $\tau \mapsto -\tau$ is a rotation by π/M, so H is even in τ, and

$$H - H_0 = h_2\tau^2 + h_4\tau^4 + O(\tau^6), \qquad h_2 = -\tfrac{M}{4}(1+b)^2\big[M^2(1+b)^2 - (4M-2)(1+b^2)\big],$$

$$h_4 = \tfrac{M(1+b)^4}{96}\big[M^4(1+b)^4 - 16M^2(1+b)(1+b^3) + 24(2M-1)(1+b^4)\big].$$

$h_2 = 0$ is exactly (2.2) for even n. Since $\varepsilon = \frac{\sqrt{n}}{2}(1+b)^2\tau + O(\tau^2)$, $c_4 = h_4\,(2/(1+b)^2)^4/n^2$. Reducing modulo (2.2) gives the closed form; every factor is positive for $m \ge 2$. □

[Proved] (symmetric criticality and the identification $f = H|_{\text{curve}}$). [Lean], from the Hamiltonian (2.1) as a function of the n labelled points: `Conj13.EvenConfig`: `Ham_stag_sub` (product formula); `Conj13.EvenQuartic`: `J3_curve` (the curve lies in $J_3^{-1}(\mu_3)$), `Jp_stag` ($J_1 = J_2 = 0$ on it), `epsK_stag` (ε and the gauge), `Ham_curve_jet` (the jet with these h_2, h_4), `c4even_limit_at_bn` ($(H - H_0)/\varepsilon^4 \to c_4$ along the curve at b_n, for every $M \geq 2$); `Conj13.Algebra`: `h2_eq_zero_iff`, `c4even_reduction`, `c4_even_at_bn`, `c4even_pos`. Not in Lean: that the curve carries the minimiser (symmetric criticality). [Numerical] `code/c4_levelset.py` minimises H on the level set directly (80-digit Newton on the full KKT system, symmetric start) for $n = 4, \ldots, 11$.

5 Odd n

Theorem 5.1. *For odd $n \geq 5$ at $b = b_n$,*

$$c_4 = \frac{(n-1)(n-3)(n^2 - 9n + 38)}{128\, n} > 0$$

($n^2 - 9n + 38$ has discriminant -71). For example $c_4 = \frac{9}{40}, \frac{9}{14}, \frac{19}{12}, \frac{75}{22}, \frac{675}{104}$ for $n = 5, 7, 9, 11, 13$.

Proof. Canonical coordinates. Put $w = |z|^2/\sigma$ and $\vartheta = \arg z$. Then $\Omega = \sum_k dw_k \wedge d\vartheta_k$, $J_3 = \sum_k w_k$ is linear, $J_+ = \sum_k Q(w_k) e^{i\vartheta_k}$ with $Q = \sqrt{w(1 - \lambda w)}$, and

$$\frac{|z_i - z_j|^2}{\sigma_i \sigma_j} = w_i + w_j - 2\lambda w_i w_j - 2Q(w_i)Q(w_j)\cos(\vartheta_i - \vartheta_j),$$

so H has no one-body term. The ring is $w_k = w_0 = 1/(1 + \lambda)$, $\vartheta_k = \phi_k$.

Ansatz. Write $w_k = w_0 + \delta_k$, $\vartheta_k = \phi_k + \psi_k$. On $J^{-1}(\mu)$, $\sum_k \delta_k = 0$ *exactly*; fix the gauge by $\sum_k \psi_k = 0$. By Lemmas 3.2 and 3.3 we may compute along the kernel direction $\delta_k = A\cos(l\phi_k)$. Since $2l \equiv -1 \pmod n$, products of two kernel modes lie in Fourier modes 0 and ± 1. So in the slaved displacement $y^*(a) = O(A^2)$, only modes 0 and 1 have an A^2 part: for $m \geq 2$ there is no cubic coupling $a^2 y_m$, and the corresponding component of y^* is $O(A^3)$. Mode 0 is excluded exactly (radial: $\sum \delta_k = 0$; tangential: gauge). The reflection $k \mapsto -k$ fixes $\cos(l\phi_k)$, so by uniqueness the mode-1 part is reflection-symmetric:

$$\delta_k = A\cos(l\phi_k) + A^2 p \cos\phi_k, \qquad \psi_k = A^2 s \sin\phi_k.$$

Corrections of order A^3 do not affect the A^4 coefficient. The truncated ansatz satisfies $J_3 = \mu_3$ exactly but $J_+ = 0$ only to order A^2 (for $n = 5$, where $3l \equiv 1 \pmod n$, J_+ has a genuine A^3 term in mode 1), so the constrained point differs from it by a correction $c = O(A^3)$ (in mode 1 to restore $J_+ = 0$, with $\sum_k \delta_k = 0$ kept, and in the slaved modes $m \geq 2$). Work with F, which equals $H - \omega_0 \mu_3$ at both points because $J_3 = \mu_3$ at both. Then $F(q + c) - F(q) = dF(q)c + O(|c|^2)$, and $dF(q) = d^2F(x_0)(Ae + A^2 y_2) + O(A^2)$, where $d^2F(x_0)e = 0$ (e spans an eigenvector of the block-diagonal $d^2F(x_0)$ with eigenvalue 0). So $dF(q)c = O(A^2) \cdot O(A^3) = O(A^5)$ and $F(q + c) - F(q) = O(A^5)$. Hence the A^4 coefficient of g is the minimum, over (p, s) satisfying the order-A^2 part of $J_1 = 0$, of the A^4 coefficient of H along the ansatz.

Constraint. $J_2 = 0$ by reflection symmetry. At order A^2, $J_1 = 0$ reads

$$\tfrac{n}{2} A^2 \big[Q'(w_0)p - Q(w_0)s + \tfrac{1}{4}Q''(w_0)\big] = 0, \qquad Q(w_0) = \tfrac{1}{1+\lambda},\ Q'(w_0) = \tfrac{1-\lambda}{2},\ Q''(w_0) = -\tfrac{(1+\lambda)^3}{4},$$

using $\sum_k \cos^2(l\phi_k)\cos\phi_k = n/4$. The inhomogeneous term $\frac{1}{4}Q''$ is the curvature of the level set; in these coordinates it sits in J_1.

n	b_n	geom.	c_4	decimal	n	b_n	geom.	c_4	decimal
4	0.2679	S^2	1/4	0.2500	15	−0.2679	H^2	56/5	11.2000
5	0.1716	S^2	9/40	0.2250	16	−0.2876	H^2	1141/80	14.2625
6	0.0557	S^2	23/30	0.7667	17	−0.3033	H^2	609/34	17.9118
7	0	$\mathbb{R}^2$	9/14	0.6429	18	−0.3198	H^2	3190/153	20.8497
8	−0.0627	H^2	95/56	1.6964	19	−1/3	H^2	27	27.0000
9	−0.1010	H^2	19/12	1.5833	20	−0.3474	H^2	11139/380	29.3132
10	−1/7	H^2	49/15	3.2667	21	−0.3592	H^2	2175/56	38.8393
11	−0.1716	H^2	75/22	3.4091	22	−0.3715	H^2	6145/154	39.9026
12	−0.2024	H^2	755/132	5.7197	23	−0.3820	H^2	2475/46	53.8043
13	−0.2251	H^2	675/104	6.4904	24	−0.3928	H^2	29183/552	52.8678
14	−0.2493	H^2	1693/182	9.3022	25	−0.4021	H^2	7227/100	72.2700

Table 1: The level-set quartic c_4 (Definition 3.1, H as in (2.1), $r_0 = 1$). [Lean] `Conj13.Algebra.table_values` (closed forms at $n = 4, \dots, 25$, exact over $\mathbb{Q}$) and `c4_at_bn` (the defining expression at the irrational b_n equals the closed form, all $n \geq 4$). [Numerical (exact modular)] `code/c4_table_modular_check.py`: an independent order-by-order KKT expansion of the level-set minimisation in Cartesian coordinates, with all $2n$ coordinates free and no ansatz, run exactly over $\mathrm{GF}(p)$ for three primes $p \approx 2^{61}$ at both conjugates of b_n, reproduces every entry, confirms that the kernel eigenvalue and the cubic term vanish, and rejects a perturbed value. (Agreement modulo three large primes is strong evidence for, not a proof of, equality over $\mathbb{Q}$; the proof is Theorems 4.1 and 5.1.) [Numerical] `code/c4_levelset.py`: direct 80-digit level-set minimisation for $n = 4, \dots, 11$ agrees to 10^{-16} or better.

Expansion. Along the ansatz, $H - H_0 = f_2 A^2 + E(p, s)A^4 + O(A^5)$ with no A or A^3 term, where

$$f_2 = -\tfrac{n(n-1)(1+\lambda)^2}{64}\big[(n-7)(1+\lambda^2) + 2(n+1)\lambda\big]$$

vanishes exactly at b_n (for odd n, (2.2) reads $(1+\lambda^2)/(1+\lambda)^2 = (n+1)/8$), and $E = E_{00} + E_{10}p + E_{01}s + E_{20}p^2 + E_{02}s^2$ with the coefficients of Appendix A. The sums over vortex pairs that occur reduce to the Fejér identity (2.3) and its second-order analogue. Eliminating s by the constraint leaves a quadratic in p with leading coefficient $a_2 = \frac{1}{8}n(n-1)(1-\lambda)^2(1+\lambda)^2 > 0$. The normalisation $\varepsilon \leftrightarrow A$ is $A = \varepsilon(2/n)^{1/2}\, 2/(1+\lambda)^2 + O(\varepsilon^2)$, so $c_4 = \frac{64}{n^2(1+\lambda)^8} \min_p E(p, s(p))$. Reducing modulo the threshold relation gives the stated formula (including $n = 7$, $\lambda = 0$). □

[Proved] (the reduction to the ansatz: which modes are slaved at order A^2, the reflection symmetry, the irrelevance of $O(A^3)$ corrections). [Lean], from the Hamiltonian as a function of the n labelled points: `Conj13.Odd4Hamiltonian`: `Ham_oddAnsatz_jet` (the full fourth-order jet along the ansatz, with these f_2 and E_{ij}, no A and no A^3 term, all odd $n \geq 5$, $\lambda \in (-1, 1)$); `Conj13.Odd4Fejer`: `fejer1`, `fejer2` (first- and second-order Fejér identities); `Conj13.OddConstraint`: `J1_oddAnsatz_jet` (the constraint); `Conj13.OddQuadratic`: `epsOdd_hasDerivAt`, `oddNorm_pow4` (the normalisation); `Conj13.Algebra`: `Q_values`, `sum_cos_sq_mul_cos`, `f2_eq_zero_iff`, `a2_pos`, `c4odd_reduction` (the constrained minimum exists and equals the closed form, for all real $n > 1$ including $n = 7$), `c4_odd_at_bn`, `c4odd_pos`, `odd_quadratic_disc`.

Remark 5.2 ($n = 3$). For $n = 3$, (2.2) has its only root in $(-1, 1]$ at $b = 1$, the equator, where $G_\mu = SO(3)$, the critical mode is $m = 1$ and the analysis above does not apply. The odd formula gives 0 at $n = 3$ (`Conj13.Symmetry`: `c4odd_three`), consistent with this. MT state that the 3-ring is always nonlinearly stable [8, p. 15]; we do not treat $n = 3$.

6 Comparison with Montaldi–Tokieda's even-n computation

MT's argument for even n [8, §3.5, pp. 19–21] has three steps. Their Lemma 8 is a criterion for a degenerate critical point of a function f to be a strict minimum: $d^3f(a,a,a)=0$ and $d^4f(a,a,a,a)+6\,d^3f(a,a,y)+3\,d^2f(y,y)>0$ for $a\in\ker d^2f\setminus 0$ and all y (MT write B,C,D for d^2f,d^3f,d^4f and b for y; we change letters to avoid clashes with (6.1) and the curvature parameter). Their Proposition 9 applies it to the augmented Hamiltonian $\hat{H}=F$ restricted to the symplectic slice N, a *linear* complement of the group directions in $\ker dJ(x_0)$, on the grounds that "the hypotheses of Lemma 8 applied on M_μ are equivalent to the same hypotheses applied on N". Their Lemma 10 shows $d^3F(a,a,y)=0$ for $a\in V_{n/2}$, $y\in N$, so the criterion reduces to $d^4F(a^4)>0$, which they evaluate as the t^4 coefficient of F along $z_j=(r_0+(-1)^jt)e^{2\pi ij/n}$.

The equivalence in Proposition 9 is correct for the 2-jet, but not for the 4-jet at a degenerate point. The reduced Hamiltonian pulled back to N by a lift $\nu\mapsto x_0+\nu+c(\nu)$, $c(\nu)=O(|\nu|^2)$ transverse to $\ker dJ$, is $F(x_0+\nu+c(\nu))$, not $F(x_0+\nu)$; the difference contains $\frac{1}{2}d^3F(\nu,\nu,c)+\frac{1}{2}d^2F(c,c)$, which is of order $|\nu|^4$ and nonzero when c has a component along radial V_0 (Remark 3.4). In Lemma 8 terms, the y that matters is $c(a)$, which is forced by the constraint, lies outside N, and is not killed by Lemma 10.

To quantify this, restrict to the $D_{n/2}$-invariant family $\rho_\pm=1+u\pm v$ of two staggered $n/2$-gons (it contains both MT's line, $u=0$, and the level-set curve of Section 4). By the symmetry $v\mapsto -v$,

$$F-F_0=v_2v^2+Au^2+Buv^2+Dv^4+\dots, \tag{6.1}$$

where the omitted terms are of order ≥ 5 when $u=O(v^2)$.

Proposition 6.1. *For even $n=2M$ and $b\in(-1,1)$, with $\vartheta(b)=M^2(1+b)^2-(4M-2)(1+b^2)$ (so $\vartheta=0$ is (2.2)):*

1. *$v_2=-M\vartheta/(1+b)^2$, and*
$$D=\frac{M\big[(M^4-4M^2)(1+b)^4+3(2M-1)(9b^4-14b^3+26b^2+2b+1)\big]}{6(1+b)^4}.$$
2. *The other coefficients are*
$$A=\frac{2M(2M-1)(1+b^2)}{(1+b)^2},\qquad B=\frac{2M\big[M^2(1+b)^3-(2M-1)(5b^3-3b^2+9b+1)\big]}{(1+b)^3},$$
and on the level set $J_3=\mu_3$ one has $u=u_2v^2+O(v^4)$ with $u_2=\dfrac{3b-1}{2(1+b)}$.
3. *The quartic along the level set is $D+Bu_2+Au_2^2$, and at $b=b_n$, $D+Bu_2+Au_2^2=n^2c_4$ with c_4 as in Theorem 4.1. Along MT's line it is D, and $\varepsilon=\sqrt{n}\,v$ exactly, so the line gives $(F-F_0)/\varepsilon^4\to D/n^2$.*

[Lean] `Conj13.MTLine`: `Fline_jet` (item 1, derived from the Hamiltonian along the line: $F(\mathrm{line}(t))-F_0-v_2t^2-Dt^4=O(t^6)$ for $|b|<1$), `v2_eq`, `Dline_eq`, `epsK_line`, `Fline_limit`; `bending_identity` (exact identity $D+Bu_2+Au_2^2-n^2c_4=\vartheta\cdot q/(2(2M-1)(1+b)^4)$, q explicit) and `bending_at_bn` (item 3 at b_n, every $M\geq 2$). Item 2, also from the Hamiltonian: `Conj13.MTFamily`: `Ffam_jet` (on every curve $u=\kappa v^2$, $F-F_0=v_2v^2+(A\kappa^2+B\kappa+D)v^4+O(v^6)$, for $|b|<1$; a quadratic in κ determines A, B, D), `c2fam_eq`, `c4fam_eq`, `J3fam_jet` and `gc2_eq_zero_iff` ($J_3-\mu_3=M\mathcal{G}_2(\kappa)v^2+O(v^4)$ with $\mathcal{G}_2(\kappa)=0\Leftrightarrow\kappa=u_2$), `level_quartic_at_bn`. [Certified] cross-check by exact symbolic differentiation (`code/mt_line_vs_levelset.py`).

n	b_n	u_2	D (MT line)	$Bu_2 + Au_2^2$	$D + Bu_2 + Au_2^2$	n^2c_4
4	0.2679	−0.0774	3.6906	0.3094	4	4
6	0.0557	−0.3944	29.6837	−2.0837	27.6	138/5
8	−0.0627	−0.6339	145.7882	−37.2168	108.5714	760/7
10	−1/7	−5/6	491.25	−164.5833	326.6667	980/3
12	−0.2024	−1.0076	1298.0966	−474.4603	823.6364	9060/11
20	−0.3474	−1.5647	18179.8554	−6454.5923	11725.2632	222780/19

Table 2: MT's straight-line quartic D against the level-set quartic n^2c_4, in the variable $v = (\rho_+ - \rho_-)/2$ (MT's t, with $r_0 = 1$ and H as in (2.1), i.e. 4π times MT's f). [Certified] `code/mt_line_vs_levelset.py` (exact symbolic in M, b; values at b_n; D recomputed independently from the vortex positions at 50 digits).

Corollary 6.2 ($n = 4$)**.** *At $n = 4$, $b_4 = 2 - \sqrt{3}$: $D = 6 - \frac{4}{3}\sqrt{3} \approx 3.6906$, whereas $n^2c_4 = 4$. Along MT's line $(F - F_0)/\varepsilon^4 \to (6 - \frac{4}{3}\sqrt{3})/16 \approx 0.2307$; on the level set the limit is $c_4 = 1/4$.*

[Lean] `Conj13.MTLine`: `n4_line_value`, `n4_line_ne`, `n4_line_limit`; the level-set limit is `Conj13.EvenQuartic`: `c4even_limit_at_bn` with $M = 2$.

Discussion. Both quartics are positive for every n in Table 2, so MT's conclusion for even n is correct; but the quantity they show to be positive is not the reduced quartic, and the bending term can have either sign (positive at $n = 4$, negative for $n \geq 6$ in the table). For example, at $n = 4$ the straight line underestimates the reduced quartic by 8%, and at $n = 20$ it overestimates it by 55%. A positive line value would not in general imply a positive reduced quartic.

Two further observations, stated with less confidence. (i) MT print the t^4 coefficient as $\frac{n}{768\pi r_0^4 \sigma^4} T$ with an explicit bracket T. Expanding the function f they display, we find ([Certified], same script) $T_{\text{direct}} = n^2(n^2 - 16)(1 + \lambda)^4 + 48(n - 1)(9\lambda^4 - 14\lambda^3 + 26\lambda^2 + 2\lambda + 1)$ at $r_0 = 1$, and $T_{\text{printed}} - T_{\text{direct}} = 72(n - 1)\lambda^2(\lambda^2 - 6\lambda + 1)$. MT's printed closed forms for $T_1 + T_2$ and $T_1 T_2$ agree with their printed T. At $n = 4$ the printed T gives 3.6236 instead of 3.6906. We have not determined whether this is a typographical or an algebraic slip; it does not affect the sign. (ii) MT's odd-n criterion $\beta_{\text{MT}}\delta_{\text{MT}} > \gamma_{\text{MT}}^2$ [8, (3.9)] also minimises over $b \in V_1' \subset N$, i.e. on the linear slice. The minimum of F on the linear slice $x_0 + \ker dJ(x_0)$ (with the kernel coordinate fixed) differs from c_4: in our normalisation it is 0.2307, 2.2779, 2.5781, 10.9142 against $c_4 = 0.25$, 1.6964, 1.5833, 6.4904 for $n = 4, 8, 9, 13$ ([Numerical] `code/c4_slice.py`). We have not recomputed MT's $\beta_{\text{MT}}, \gamma_{\text{MT}}, \delta_{\text{MT}}$ themselves, so we do not claim that their odd numbers are those linear-slice values.

7 Nonlinear stability at a degenerate minimum

That a strict local minimum of the reduced Hamiltonian implies Lyapunov stability modulo G_μ is the energy–momentum method; it requires no definiteness of the Hessian. See Patrick [11] for the dynamic interpretation on the reduced space, Ortega and Ratiu [9, 10] and Patrick, Roberts and Wulff [12] for general statements (including noncompact groups, with G_μ compact), and Montaldi [7]. We claim no novelty here. Because the plane's momentum map is not equivariant and we want the result for perturbations of arbitrary momentum, we record a short self-contained version adapted to the present situation; its abstract core is checked in Lean.

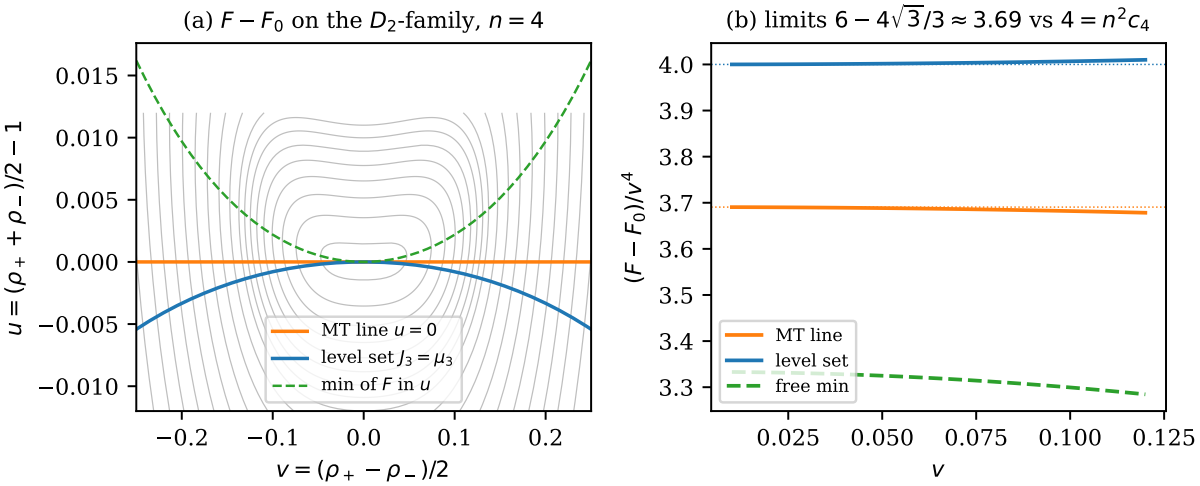


Figure 1: $n = 4$ at $b_4 = 2 - \sqrt{3}$, on the family $\rho_\pm = 1 + u \pm v$. (a) Contours of $F - F_0$ with MT's line $u = 0$, the momentum level set $J_3 = \mu_3$ (which bends to $u \approx u_2 v^2$), and, dashed, the curve of minima of F in u at fixed v (no momentum constraint). (b) $(F - F_0)/v^4$ along the three curves. The line tends to $D = 6 - \frac{4}{3}\sqrt{3} \approx 3.691$, the level set to $n^2 c_4 = 4$, and the free minimum to $D - B^2/(4A) = 10/3$. [Numerical] `code/make_figures.py`.

Hypotheses. (P, Ω) is symplectic, H smooth, a Lie group G acts symplectically preserving H, and J_1, J_2, J_3 are Hamiltonians for generators ξ_1, ξ_2, ξ_3, with ξ_3 generating a circle action K and $\mathrm{span}(\xi_1, \xi_2)$ invariant under Ad_K. x_0 is a relative equilibrium: $dF(x_0) = 0$ for $F = H - \omega_0 J_3$. Assume:

(A1) $J_1(x_0) = J_2(x_0) = 0$, K rotates (J_1, J_2), and $\{J_1, J_2\}(x_0) \neq 0$;

(A2) K acts freely near x_0;

(A3) there is a smooth curve $j \mapsto x_j$ of relative equilibria through x_0 with $J_1 = J_2 = 0$ and $J_3(x_j) = \mu_3 + j$;

(A4) on $M_\mu = J^{-1}(\mu)/K$, h_μ has positive semidefinite Hessian at $[x_0]$, definite on a complement E of its kernel Z_0, and $g(a) = \min_y h_\mu(a, y) - h_\mu(0)$ has zero cubic term and positive definite quartic term.

Theorem 7.1. *Under (A1)–(A4), $K \cdot x_0$ is Lyapunov stable in P, for perturbations of arbitrary momentum.*

Proof. Lemma A (cross-section). By (A1), $S = \{J_1 = J_2 = 0\}$ is near x_0 a K-invariant symplectic submanifold of codimension 2 (dJ_1, dJ_2 are independent and $\Omega(X_{J_1}, X_{J_2}) \neq 0$), invariant under the flow, on which the flow is that of $H|_S$. The implicit function theorem applied to $(x, s) \mapsto (J_1, J_2)(\exp(s_1\xi_1 + s_2\xi_2)x)$, whose s-derivative at $(x_0, 0)$ is invertible by (A1), gives a continuous, K-equivariant $\Lambda : U \to G$ on a K-invariant neighbourhood U of $K \cdot x_0$ with $\Lambda(x)x \in S$ and $\Lambda = e$ on $K \cdot x_0$ (uniqueness of the small solution makes the local definitions on K-translates agree).

Lemma B1. In coordinates $(a, y) \in Z_0 \times E$ on M_μ, there are $c_1, \varrho > 0$ with $h_\mu(a, y) - h_\mu(0) \geq c_1(|a|^4 + |y|^2)$ for $|(a, y)| < \varrho$. Indeed $h_\mu(a, y) - g(a) - h_\mu(0) \geq c|y - y^*(a)|^2$ by Taylor's theorem in y, $g(a) \geq c'|a|^4$ by (A4), and $|y|^2 \leq 2|y - y^*(a)|^2 + 2L^2|a|^4$.

Lemma B2. By (A2) S/K is a manifold near $[x_0]$. The function J_3 is K-invariant and descends to it with nonzero differential: on the symplectic manifold S its Hamiltonian vector field is the generator of K, which is tangent to S and nonzero by (A2), so $dJ_3|_S \neq 0$, and $dJ_3|_S$ vanishes on the orbit direction. Hence J_3 is a submersion on S/K near $[x_0]$, and its level sets near $[x_0]$ are the reduced spaces. Choose coordinates (j, w), $j = J_3 - \mu_3$, with $\{w = 0\}$ the curve of (A3) (transverse to the levels since $J_3(x_j) = \mu_3 + j$) and $\{j = 0\} = M_\mu$ carrying the coordinates of B1. Let $\Phi(j, w) = H(j, w) - H(j, 0)$. Since x_j is critical for H on its level, $\Phi(j, 0) = 0$ and $\partial_w \Phi(j, 0) = 0$, hence $|\partial_j \Phi(0, w)| \leq L'|w|^2$ and $\Phi(j, w) \geq \Phi(0, w) - L'|j|(|a|^2 + |y|^2) - C_2 j^2$. With B1 and $L'|j||a|^2 \leq \frac{c_1}{2}|a|^4 + \frac{L'^2}{2c_1} j^2$, $\Phi \geq \frac{c_1}{2}(|a|^4 + |y|^2) - Cj^2$ near 0.

Lemma B3. $V = H - \omega_0 J_3 + c_\mathrm{p}(J_3 - \mu_3)^2$ is conserved and K-invariant on S. Writing $V - V_0 = \Phi + \psi(j) + c_\mathrm{p} j^2$ with $\psi(j) = H(x_j) - H(x_0) - \omega_0 j = O(j^2)$ (because $\frac{d}{dj} H(x_j) = \omega(j)$), $V - V_0 \geq \frac{c_1}{2}(|a|^4 + |y|^2) + (c_\mathrm{p} - C - C_3) j^2$, a strict minimum on S/K at $[x_0]$ for c_p large.

Lemma B4. A conserved continuous function with a strict minimum modulo the compact orbit $K \cdot x_0$ gives Lyapunov stability of $K \cdot x_0$ for the flow on S; solutions stay in a compact collision-free set and so exist for all time.

Conclusion. For $x(0)$ near $K \cdot x_0$ let $g_0 = \Lambda(x(0))$. Since G commutes with the flow, $\tilde{x}(t) = g_0 x(t)$ is the solution through $g_0 x(0) \in S$, which stays near $K \cdot x_0$ by B4, and $\operatorname{dist}(x(t), K \cdot x_0) \leq \operatorname{dist}(\tilde{x}(t), K \cdot x_0) + \sup d(g_0^{-1} z, z)$, the supremum over z in a compact neighbourhood of $K \cdot x_0$, which tends to 0 as $g_0 \to e$. □

[Proved]. [Lean] for the abstract core, in a metric-space setting with the chart, slice and slaving map as hypotheses: `Conj13.LyapunovAnalytic`: `lemma_B1` (from the existence of y^* and the slaving inequality), `lemma_B2`, `lemma_B3`; `Conj13.LyapunovAbstract`: `lyapunov_on_section` (B4) and `stable_modulo_orbit` (the conclusion, for a topological group acting continuously, given a map Λ as in Lemma A and G-equivariant trajectories). These are stated for trajectories defined on a time interval $I(x) \ni 0$, which is the situation of the vortex flow, whose solutions are a priori only local; that they exist for all time is the last step of B4, by hand (they stay in a compact collision-free set). Not in Lean: Lemma A itself, the manifold structure of S/K and the chart, the existence of y^*, global existence, and the vortex flow.

Remark 7.2. The penalty term in B3 is needed in every geometry except the plane. With λ fixed, the ring at momentum $\mu_3 + j$ has radius r with $J_3 = nr^2/(1 + \lambda r^2)$ increasing in r, so its parameter $b = \lambda r^2$ moves away from b_n: on the sphere ($n = 4, 5, 6$, $\lambda > 0$) $b > b_n$ for $j > 0$, and on the hyperbolic plane ($n \geq 8$, $\lambda < 0$) $b > b_n$ for $j < 0$. Since $C_1(b) = 2(n-1)(1 + b^2)/(1 + b)^2$ is strictly decreasing on $(-1, 1)$, the kernel eigenvalue $\alpha_{\lfloor n/2 \rfloor} = C_1(b) - \lfloor n^2/4 \rfloor$ is then negative and the ring is a saddle of the reduced Hamiltonian at that momentum ([Proved], from Proposition 2.3); the bifurcated minima have depth $O(j^2)$. So the reduced Hamiltonians at nearby momenta do not have uniform strict minima, and B2 only bounds how far below zero Φ can go. In the plane ($n = 7$) $b = 0$ for every r, the ring stays at the threshold, and the penalty is harmless but not needed.

Remark 7.3 (The free minimum gives no certificate from D). On the family (6.1), minimising F over u with no momentum constraint gives the quartic $D - B^2/(4A)$ (Figure 1(a), dashed). It is a lower bound for both MT's line value D and the level-set value $D + Bu_2 + Au_2^2$, since $A > 0$. At $n = 4$ it is $10/3 \approx 3.333$ (against $D \approx 3.691$ and 4), at $n = 8$ it is $751/7 \approx 107.29$ (against 145.79 and 108.57) ([Certified] `code/mt_line_vs_levelset.py`). Positivity of D therefore bounds nothing from below; positivity of the free minimum would, within this family.

Proof of Theorem 1.1. We check (A1)–(A4) for the ring at b_n, $n \geq 4$, in all three geometries. (A1): $J_+(x_0) = 0$ by $\mathbb{Z}_n$-symmetry, a rotation by ψ multiplies J_+ by $e^{i\psi}$, $\{J_1, J_2\}(x_0) = n(1 -$

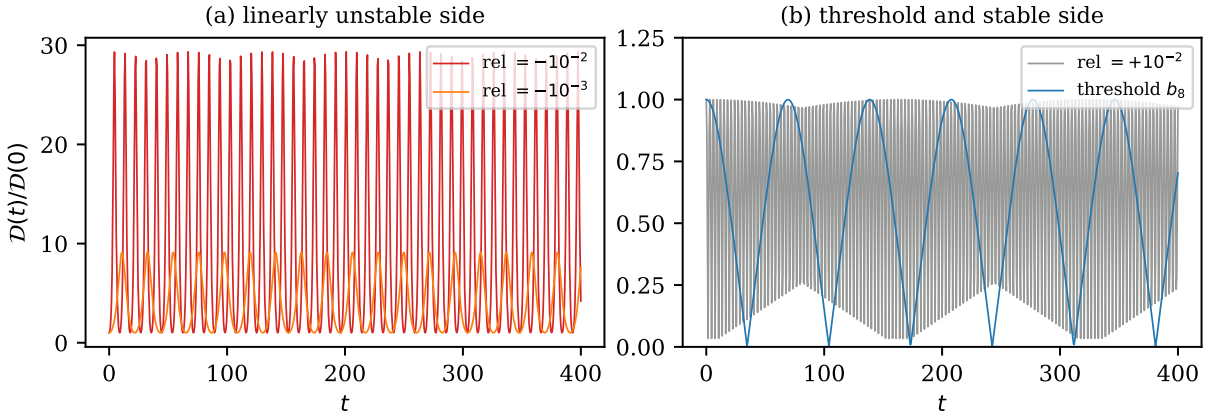


Figure 2: $n = 8$, hyperbolic plane: $\mathcal{D}(t)/\mathcal{D}(0)$ for a critical-mode perturbation $\delta = 10^{-3}$ of the ring at relative offset rel from the threshold. (a) Unstable side: the orbit wanders near the bifurcated pair. (b) Threshold (blue) and stable side (grey): the deviation never exceeds its initial value. [Numerical].

$b_n)/(2(1 + b_n)) \neq 0$ since $|b_n| < 1$ (Proposition 2.2), and $\mathrm{span}(\xi_1, \xi_2)$ is Ad_K-invariant in $\mathfrak{so}(3)$, $\mathfrak{se}(2)$ and $\mathfrak{sl}(2, \mathbb{R})$. (A2): a nontrivial rotation moves a labelled ring. (A3): the concentric rings of radius r are relative equilibria with $J_+ = 0$ and $J_3 = nr^2/(1 + \lambda r^2)$ strictly increasing in r. (A4): Proposition 2.4, Lemma 3.3, and $c_4 > 0$ by Theorems 4.1 and 5.1; the quartic term of g is $c_4|a|^4$ in the chart of Definition 3.1, and by Lemma 3.2 its positivity is chart-independent. Theorem 7.1 gives Theorem 1.1. Since K is compact, K-invariant neighbourhoods of $K \cdot x_0$ are cofinal, so this is stability in MT's sense [8, p. 10]. □

[Proved] (the instantiation). The facts used are Lean-checked as cited above, except the reduced-space structure, the Poisson structure of Ω and the vortex flow, which are by hand.

8 A direct integration

As a check on the theory (not part of the proof), we integrated the $n = 8$ ring in the hyperbolic plane (Poincaré disk of curvature -1, DOP853, relative tolerance 10^{-12}, $T = 2000$). With $\zeta = \sqrt{-\lambda}\, z$, MT's chart for $\lambda < 0$ becomes the curvature -1 Poincaré disk, $\sigma = 1 - |\zeta|^2$, and Ω becomes a constant multiple of its area form; a ring of radius r_0 has $\xi := |\zeta|^2 = -\lambda r_0^2 = -b$. So the threshold is $\xi^* = -b_8 = 8 - 3\sqrt{7}$, and rel > 0 in $\xi = \xi^*(1 + \mathrm{rel})$ means $b < b_8$, the linearly stable side. We start from the ring at $\xi = \xi^*(1 + \mathrm{rel})$ plus the critical mode with relative amplitude $\delta = 10^{-3}$ (Figure 2). The diagnostic $\mathcal{D}(t) = \max_{i<j} |d_{ij}(t) - d_{ij}^{\mathrm{ring}}|$ uses hyperbolic distances and is invariant under the whole group. On the linearly unstable side the deviation grows to an amplitude that scales like $\sqrt{|\mathrm{rel}|}$ (the stable bifurcated $n/2$-gon pair of a supercritical pitchfork); at the threshold and on the stable side $\max_t \mathcal{D}(t)/\mathcal{D}(0) = 1.000$ to three decimals, with no secular growth. [Numerical]: the traces in Figure 2 are recorded outputs of the original integration (not rerun); `code/n8_integration.py` reruns the same experiment and prints the maxima.

9 Relation to prior work

Montaldi–Tokieda [8]. We quote arXiv:1210.5662v1 (the only arXiv version); we did not consult the published chapter (Springer, 2013, pp. 335–370), so page and equation numbers, the bracket T and Proposition 9 are as in the arXiv text. MT introduce the family, prove linear stability below b_n and instability above, identify the pitchfork, and pose Conjecture 13. At the threshold they treat even n by the straight-line quartic of Section 6 (with Maple closed forms for $T_1 + T_2$ and T_1T_2), $n = 5, 9, 11$ by 2–4 significant-digit evaluations of their coefficients $\beta_{\mathrm{MT}}, \gamma_{\mathrm{MT}}, \delta_{\mathrm{MT}}$, $n = 7$ by Maple, and $n = 3$ by a remark. Our symmetry argument for the absence of a cubic term on the kernel is theirs [8, Prop. 12].

The heptagon. For $n = 7$ (the plane) nonlinear stability is due to Kurakin and Yudovich [6] and Schmidt [13]; MT recover it [8, Cor. 7]. Our $n = 7$ case is a special case of the uniform argument.

Critical cases elsewhere. Kurakin [5] carries out a critical-case Lyapunov analysis for the regular n-gon inside a circular domain. For the sphere, Borisov and Kilin [1] note that the critical latitude needs a separate nonlinear analysis, and Kurakin [3, 4] gives conditions for nonlinear stability of regular vortex systems on the sphere. Constantineau, García-Azpeitia, García-Naranjo and Lessard [2] give computer-assisted proofs of the stability of branches of relative equilibria on the sphere bifurcating from the rings, away from the degenerate point, not at the threshold itself.

Energy–momentum method. The step from a strict minimum of the reduced Hamiltonian to Lyapunov stability modulo G_μ is standard [11, 9, 10, 12, 7]; Section 7 only adapts it.

What is new here. To our knowledge (we have not found these in the literature): (i) a proof of Conjecture 13 uniform in $n \geq 4$, in particular for all odd $n \geq 13$; (ii) the exact closed forms of the level-set quartic c_4 for all n; (iii) the observation that MT's even-n quartic is taken off the momentum level set and differs from the reduced quartic, with the exact correction (Proposition 6.1); (iv) machine-checked proofs of the algebraic and series steps.

10 Status and limitations

Review status. No human specialist has refereed this argument. It has been checked by computer, including independent re-derivations (Lean, exact symbolic and modular computation, high-precision numerics), and by AI-assisted adversarial review. Its claims should be treated accordingly.

What Lean checks. Library `Conj13` in `lean/Conj13/` of the accompanying repository (Lean 4, Mathlib v4.34.1), with no `sorry` and only the standard axioms `propext`, `Classical.choice`, `Quot.sound` (`outputs/lean_axioms.out`):

- the threshold b_n: closed form, uniqueness, sign pattern (Proposition 2.2);
- the spectral picture: from the Hamiltonian, that the ring is critical exactly for ω_0 and that the Hessian is block-diagonal over Fourier modes with blocks $\mathrm{diag}(\alpha_m, \beta_m)$, for every $b > -1$; the Fejér identity for all n, the factorisation and sign of α_m, the V_1' eigenvalues (Propositions 2.3, 2.4);

- the partial derivatives and the value of $\{J_1, J_2\}$ (Section 2.4);
- the invariant-monomial arithmetic of Lemma 3.3;
- for even n, starting from the Hamiltonian of n labelled points: the product formula, that the staggered curve lies in $J^{-1}(\mu)$ and in the gauge, its jet, the normalisation, and the limit $(H - H_0)/\varepsilon^4 \to c_4$ at b_n for every n;
- for odd n, again from the Hamiltonian: the full fourth-order jet along the ansatz (including the pair sums via first- and second-order Fejér identities and the absence of the A^3 term), the J_1 constraint, the normalisation, and the constrained minimisation and its reduction modulo the threshold;
- the closed forms of c_4 at the true b_n, their positivity for every $n \geq 4$, and Table 1;
- the straight-line jet of F and the jets on every curve $u = \kappa v^2$ (both from the Hamiltonian, which identifies A, B, D and u_2), the bending identity, and the $n = 4$ comparison (Proposition 6.1, Corollary 6.2);
- the abstract Lyapunov argument (Lemmas B1–B4 and the conclusion of Theorem 7.1), with the slice, the chart and the slaving map as hypotheses.

What is by hand. The reduced space and the restriction of the Hessian to $\ker dJ$ (Proposition 2.4; the momentum differentials and V_1'); the identification of the polarised second derivatives with the Hessian matrix; the Poisson structure of Ω (taken as a definition in the bracket computation); the chart-independence and symmetry lemmas; symmetric criticality for even n; the completeness of the odd ansatz (which modes are slaved at order A^2, the reflection symmetry, the irrelevance of $O(A^3)$ corrections) and the identification of the constrained minimum with the quartic of g; Lemma A, the manifold chart on S/K and the slaving map; and the instantiation of Theorem 7.1 for the vortex flow (existence of solutions, conservation laws). These steps are supported independently by the exact GF(p) Cartesian KKT computation (no ansatz, no symmetry reduction) and by the high-precision numerics, but those are not proofs.

Scope. The result concerns MT's family (2.1), i.e. the "sinh" interaction on the hyperbolic plane. The same method extends to other hyperbolic interactions (for instance the image or "tanh" kernel) and to interpolating families; we do not treat them here. We do not address the bifurcated branches, KAM-type questions, or $n = 3$.

Acknowledgement and disclosure of AI assistance

This work was developed with substantial assistance from an AI system (Claude, Anthropic), covering the derivations, the Lean formalisation, the computer-algebra and numerical code, and the drafting of this manuscript; the author directed the work and takes responsibility for the content. No human expert has yet checked the mathematics, and corrections are welcome. We thank James Montaldi and Tadashi Tokieda, whose paper posed the problem and supplied the framework used here.

A Coefficients for odd n

Along the ansatz of Theorem 5.1, $H - H_0 = f_2 A^2 + E(p,s)A^4 + O(A^5)$ with $E = E_{00} + E_{10}p + E_{01}s + E_{20}p^2 + E_{02}s^2$ (no ps term) and

$$E_{20} = \tfrac{1}{16}n(n-1)(1-\lambda)^2(1+\lambda)^2, \qquad E_{02} = \tfrac{1}{4}n(n-1),$$
$$E_{10} = \tfrac{1}{64}n(n-1)(1-\lambda)(1+\lambda)^3\big[(n-5)(1+\lambda^2)+2(n-1)\lambda\big], \qquad E_{01} = \tfrac{1}{64}n(n-1)^2(1+\lambda)^4,$$

$$E_{00} = \tfrac{n(n-1)(1+\lambda)^4}{8192}\Big[(n^3+n^2-69n+291)(1+\lambda^4)+4(n^3+n^2-21n-13)(\lambda+\lambda^3) + (6n^3+6n^2-30n+82)\lambda^2\Big].$$

The constraint is $\frac{1-\lambda}{2}p - \frac{1}{1+\lambda}s - \frac{(1+\lambda)^3}{16} = 0$. Eliminating s, the coefficient of p^2 is $a_2 = \frac{1}{8}n(n-1)(1-\lambda)^2(1+\lambda)^2$. [Lean]: `Conj13.Odd4Hamiltonian`: `Ham_oddAnsatz_jet` derives this jet, with exactly these coefficients (definitions `E00`, …, `E02`, `Efun` in `Conj13.Algebra`), from the Hamiltonian; `J1con`, `E_vertex`, `odd_min_value`, `c4odd_reduction` do the elimination.

B Expansions behind h_4 and the E_{ij}

These are the steps a reader needs to reproduce the coefficients by hand; Lean checks each of them (`Conj13.EvenCurve`: `Phi_jet`; `Conj13.Odd4Pair`: `jet_logPw`, `jet_pair`; `Conj13.Odd4Fejer`).

Even $n = 2M$. On the curve $\mathcal{G}(u_\pm) = \mathcal{G}(1) \pm \tau$ one has, with $x = (1+b)\tau$, $u_\pm = (1 \pm x)/(1 \mp bx)$ and $1 + bu_\pm = (1+b)/(1 \mp bx)$. Hence $\log u_+ + \log u_- = \log(1-x^2) - \log(1-b^2x^2)$, $\log u_+ - \log u_- = 2(\operatorname{artanh} x + \operatorname{artanh} bx)$, and the product formula of Theorem 4.1 becomes

$$H - H_0 = -\tfrac{M(2M-1)}{2}\big[\log(1-x^2)+\log(1-b^2x^2)\big] - 2M\log\cosh Y, \qquad Y = \tfrac{M}{2}\big(\operatorname{artanh} x + \operatorname{artanh} bx\big),$$

using $\log\big((u_+^{M/2}+u_-^{M/2})/2\big) = \frac{M}{4}(\log u_+ + \log u_-) + \log\cosh\big(\frac{M}{4}(\log u_+ - \log u_-)\big)$. Insert $\log(1-x^2) = -x^2 - \frac{1}{2}x^4$, $Y = c_1 x + c_3 x^3$ with $c_1 = \frac{M}{2}(1+b)$, $c_3 = \frac{M}{6}(1+b^3)$, and $\log\cosh Y = \frac{1}{2}c_1^2x^2 + (c_1c_3 - \frac{1}{12}c_1^4)x^4$, all up to $O(x^6)$. The x^2 and x^4 coefficients, multiplied by $(1+b)^2$ and $(1+b)^4$, are h_2 and h_4.

Odd n. In canonical coordinates $|z_i - z_j|^2/(\sigma_i\sigma_j) = D_{ij}$ with $D_{ij} = w_i + w_j - 2\lambda w_i w_j - 2Q_iQ_j\cos\Theta_{ij}$, $Q_k = \sqrt{P(w_k)}$, $P(w) = w(1-\lambda w)$, $\Theta_{ij} = \vartheta_j - \vartheta_i$. Along the ansatz write $w_k = w_0 + e_kA + f_kA^2$ with $e_k = \cos(l\phi_k)$, $f_k = p\cos\phi_k$, and $\Theta_{ij} = \theta + A^2 d$ with $\theta = \phi_j - \phi_i$, $d = s(\sin\phi_j - \sin\phi_i)$. Factor $D = 2Q_iQ_j(K+Y)$ with $K = 1 - \cos(\theta + A^2d)$ and $Y = (w_i + w_j - 2\lambda w_iw_j)/(2Q_iQ_j) - 1 = O(A^2)$. Then, writing $K_0 = 1 - \cos\theta$,

$$\log D - \tfrac{1}{2}\log P_i - \tfrac{1}{2}\log P_j = \log 2 + \log K_0 + \frac{K_2+Y_2}{K_0}A^2 + \frac{Y_3}{K_0}A^3 + \Big(\frac{K_4+Y_4}{K_0} - \frac{(K_2+Y_2)^2}{2K_0^2}\Big)A^4 + O(A^5),$$

with $K_2 = d\sin\theta$, $K_4 = \frac{1}{2}d^2\cos\theta$, $Y_2 = \frac{1}{8}(1+\lambda)^4(e_i-e_j)^2$,

$$Y_3 = \tfrac{1}{8}(\lambda-1)(1+\lambda)^5(e_i-e_j)^2(e_i+e_j) + \tfrac{1}{4}(1+\lambda)^4(e_i-e_j)(f_i-f_j),$$
$$\begin{aligned} Y_4 = {} & \tfrac{3}{128}(1+\lambda)^6(5\lambda^2-6\lambda+5)(e_i-e_j)^2(e_i^2+e_j^2) + \tfrac{1}{64}(1+\lambda)^6(9\lambda^2-14\lambda+9)(e_i-e_j)^2e_ie_j \\ & + \tfrac{1}{8}(\lambda-1)(1+\lambda)^5(e_i-e_j)^2(f_i+f_j) \\ & + \tfrac{1}{4}(\lambda-1)(1+\lambda)^5(e_i-e_j)(f_i-f_j)(e_i+e_j) + \tfrac{1}{8}(1+\lambda)^4(f_i-f_j)^2. \end{aligned}$$

The one-body term is $\log P(w_0+eA+fA^2) = \log w_0^2 + L_1 eA + (L_1 f + L_2 e^2)A^2 + (2L_2 ef + L_3 e^3)A^3 + (L_2 f^2 + 3L_3 e^2 f + L_4 e^4)A^4 + O(A^5)$, where $L_1 = (1+\lambda)(1-\lambda)$, $L_2 = -\frac{1}{2}(1+\lambda)^2(1+\lambda^2)$, $L_3 = \frac{1}{3}(1+\lambda)^3(1-\lambda^3)$, $L_4 = -\frac{1}{4}(1+\lambda)^4(1+\lambda^4)$ are the Taylor coefficients of $\log(1+(1+\lambda)x)+\log(1-\lambda(1+\lambda)x)$. Since each vortex lies in $n-1$ pairs,

$$H - H_0 = -\sum_{i<j} \left[(\log D - \tfrac{1}{2}\log P_i - \tfrac{1}{2}\log P_j) - (\cdot)_{A=0}\right] - \tfrac{n-1}{2}\sum_k \left[\log P_k - \log w_0^2\right].$$

Products of cosines and sines of ϕ_i, ϕ_j are reduced to single frequencies by the product-to-sum formulas; summing over i at fixed $\theta = \phi_j - \phi_i$ removes every term whose total frequency is not $\equiv 0 \pmod n$ (here $2l \equiv -1$ matters), and the remaining sums over $\theta = 2\pi m/n$ are

$$\sum_{m=1}^{n-1} \frac{\cos(q\theta_m)}{1-\cos\theta_m} = \frac{n^2-1}{6} - q(n-q),$$

$$\sum_{m=1}^{n-1} \frac{\cos(q\theta_m)}{(1-\cos\theta_m)^2} = \frac{n^4 - 30n^2q^2 + 10n^2 + 60nq^3 - 60nq - 30q^4 + 60q^2 - 11}{180}$$

for $0 \le q \le n$, together with $\sum_m \cos(q\theta_m) = n[n \mid q] - 1$. This yields f_2 and the E_{ij} of Appendix A.

References

[1] A. V. Borisov and A. A. Kilin. Stability of Thomson's configurations of vortices on a sphere. *Regular and Chaotic Dynamics*, 5(2), 2000. arXiv:nlin/0503068.

[2] Kevin Constantineau, Carlos García-Azpeitia, Luis C. García-Naranjo, and Jean-Philippe Lessard. Determination of stable branches of relative equilibria of the N-vortex problem on the sphere. *Communications in Mathematical Physics*, 406(2):Paper No. 47, 2025. arXiv:2309.04320.

[3] L. G. Kurakin. On the nonlinear stability of regular vortex polygons and polyhedrons on a sphere. *Doklady Physics*, 48(2):84–89, 2003.

[4] L. G. Kurakin. On nonlinear stability of the regular vortex systems on a sphere. *Chaos*, 14(3):592–602, 2004.

[5] L. G. Kurakin. On the stability of Thomson's vortex configurations inside a circular domain. *Regular and Chaotic Dynamics*, 15(1):40–58, 2010.

[6] L. G. Kurakin and V. I. Yudovich. The stability of stationary rotation of a regular vortex polygon. *Chaos*, 12(3):574–595, 2002.

[7] James Montaldi. Persistence and stability of relative equilibria. *Nonlinearity*, 10(2):449–466, 1997.

[8] James Montaldi and Tadashi Tokieda. Deformation of geometry and bifurcations of vortex rings. In *Recent Trends in Dynamical Systems*, volume 35 of *Springer Proceedings in Mathematics & Statistics*, pages 335–370. Springer, Basel, 2013. arXiv:1210.5662 (v1 cited for page numbers).

[9] Juan-Pablo Ortega and Tudor S. Ratiu. Stability of Hamiltonian relative equilibria. *Nonlinearity*, 12(3):693–720, 1999.

[10] Juan-Pablo Ortega and Tudor S. Ratiu. *Momentum Maps and Hamiltonian Reduction*, volume 222 of *Progress in Mathematics*. Birkhäuser, Boston, 2004.

[11] George W. Patrick. Relative equilibria in Hamiltonian systems: the dynamic interpretation of nonlinear stability on a reduced phase space. *Journal of Geometry and Physics*, 9(2):111–119, 1992.

[12] George W. Patrick, Mark Roberts, and Claudia Wulff. Stability of Poisson equilibria and Hamiltonian relative equilibria by energy methods. *Archive for Rational Mechanics and Analysis*, 174(3):301–344, 2004. arXiv:math/0201239.

[13] Dieter Schmidt. The stability of the Thomson heptagon. *Regular and Chaotic Dynamics*, 9(4):519–528, 2004.

[14] J. J. Thomson. *A Treatise on the Motion of Vortex Rings. An Essay to which the Adams Prize was Adjudged in 1882, in the University of Cambridge*. Macmillan, London, 1883.